

$$1. \quad T_s = 114 + 273.15 = 387.15 \text{ K}$$

$$\begin{aligned} \Delta H &= \frac{V_s \cdot d}{\bar{u}} \left(1.5 + 2.7 \cdot \frac{T_s - T_a}{T_s} \cdot d \right) \\ &= \frac{13.5 \times 5}{4} \times \left(1.5 + 2.7 \times \frac{99}{387.15} \times 5 \right) \\ &\approx 16.875 \times (1.5 + 3.45) \\ &\approx 83.53 \text{ m} \end{aligned}$$

$$\begin{aligned} H_e &= H_s + \Delta H \\ &= 100 \text{ m} + 83.53 \text{ m} \\ &= 183.53 \text{ m} \end{aligned}$$

答：烟气抬升高度为 83.53 m，有效源高为 183.53 m

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$$\begin{aligned} Q &= \frac{1000 \text{ cm}^3}{\text{m}^3} \times 4.1 \times 10^4 \text{ m}^3/\text{s} \times 64 \text{ g} \times 10^3 \\ &= \frac{262.4 \times 10^4}{80640} \text{ g/s} \end{aligned}$$

$$\begin{aligned} C_{\max} &= \frac{0.234 Q}{U H^2} \times \frac{\sigma_z}{\sigma_y} \\ &= \frac{0.234}{5 \text{ m/s} \times 180^2 \text{ m}^2} \times \frac{262.4 \times 10^4}{80640} \text{ g/s} \\ &= \frac{61.4016 \times 10^4}{13.06368 \times 10^9} \text{ g/m}^3 \\ &= 4.7 \times 10^{-2} \text{ mg/m}^3 \end{aligned}$$

答：该高架源在地面轴线上最大落地浓度为 $4.7 \times 10^{-2} \text{ mg/m}^3$

$$3. \quad Q = 800 \text{ kg/h} = \frac{2}{9} \times 10^3 \text{ g/s} \quad d = 3 \text{ m}$$

$$T_s = (130 + 273.15) \text{ K} = 403.15 \text{ K} \quad T_a = 283.15 \text{ K}$$

$$\bar{u} = a \cdot V_0 = 1.4 \times 4 \text{ m/s} = 5.6 \text{ m/s}$$

$$\begin{aligned} \Delta H &= \frac{V_s \cdot d}{\bar{u}} \left(1.5 + 2.7 \times \frac{T_s - T_a}{T_s} \cdot d \right) \\ &= \frac{20 \text{ m/s} \times 3 \text{ m}}{5.6 \text{ m/s}} \times \left(1.5 + 2.7 \times \frac{120}{403.15} \times 3 \right) \\ &\approx \frac{60}{5.6} \times (1.5 + 2.41) \text{ m} \end{aligned}$$

$$\begin{aligned} &= 16.07 + 25.82 \quad H_e = \Delta H + H_s = 41.89 + 60 \\ &\approx 41.89 \text{ m} \quad = 101.89 \text{ m} \end{aligned}$$

$$\sigma_z = \frac{H}{\frac{H}{15}} = \frac{101.89}{1.414} = 0.465 \rho (-1.35 + 0.793 \ln x + 0.002 \times \ln^2 x) \approx 72.06$$

$$\ln t = \ln x \quad \ln(154.96) = -1.35 + 0.793t + 0.002t^2 = 5.04$$

$$0.002t^2 + 0.793t - 6.39 = 0$$

$$t = 7.9 = \ln x$$

$$x = 2697.28 \text{ m}$$

$$\sigma_y = (-0.006 \times 7.9 + 0.108) \times 2697.28$$

$$= (-0.0474 + 0.108) \times 2697.28 \approx 163.46$$

$$C_{\max} = \frac{0.234Q}{UH^2} \times \frac{\sigma_z}{\sigma_y}$$

$$= \frac{0.234}{5.6 \text{ m/s} \times 101.89^2 \text{ m}^2} \times \frac{2}{9} \times 10^3 \text{ g/s} \times \frac{72.06}{163.46}$$

$$= \frac{33.72408 \times 10^3}{85527377.4834864} \text{ g/m}^3$$

$$= 0.00039 \text{ g/m}^3$$

$$\approx 0.39 \text{ mg/m}^3$$

答:可能出现的最大浓度约为 0.39 mg/m^3

最大浓度点距发电站的距离为 2697.28 m